

Number it: Temporal Grounding Videos like Flipping Manga

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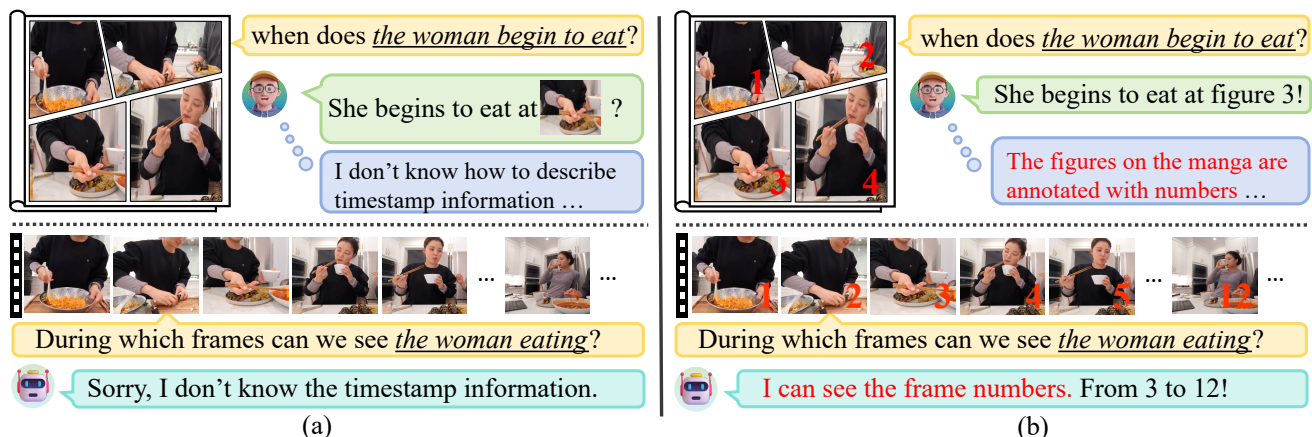


Figure 1. **Effectiveness of Adding Frame Numbers for Temporal Grounding:** (a) Without numbered images or frames, both humans and Vid-LLMs struggle to locate specific timestamps accurately. (b) Once numbered, grounding temporal cues becomes as intuitive as flipping manga, where timestamps are accessible at a glance.

Abstract

Video Large Language Models (Vid-LLMs) have made remarkable advancements in comprehending video content for QA dialogue. However, they struggle to extend this visual understanding to tasks requiring precise temporal localization, known as Video Temporal Grounding (VTG). To address this, we introduce Number-Prompt (NumPro), a novel method that empowers Vid-LLMs to bridge visual comprehension with temporal grounding by adding unique numerical identifiers to each video frame. Treating a video as a sequence of numbered frame images, NumPro transforms VTG into an intuitive process: flipping through manga panels in sequence. This allows Vid-LLMs to “read” event timelines, accurately linking visual content with cor-

responding temporal information. Our experiments demonstrate that NumPro significantly boosts VTG performance of top-tier Vid-LLMs without additional computational cost. Furthermore, fine-tuning on a NumPro-enhanced dataset defines a new state-of-the-art for VTG, surpassing previous top-performing methods by up to 6.9% in mIoU for moment retrieval and 8.5% in mAP for highlight detection. The code is available at <https://github.com/yongliang-wu/NumPro>.

1. Introduction

Imagine you are watching a cooking video, and trying to locate the exact moment when the chef stirs in the spices. While recognizing such actions is feasible, translating that visual information into precise timing, i.e., a specific second or frame number, is surprisingly difficult. This chal-

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lenge is central to the field of Video Temporal Grounding (VTG) [4, 18, 25, 36, 52, 58]. In the realm of Video Large Language Models (Vid-LLMs) [35, 43, 54, 66, 84, 89] which process videos as a sequence of frame images, the integration of VTG allows for fine-grained visual and temporal understanding and reasoning of videos, which is pivotal for developing end-to-end video dialogue systems.

Despite advances of Vid-LLMs, endowing these models with effective VTG abilities presents a unique challenge: enhancing the model’s visual recognition of an event within a video does not inherently enable it to describe when the event begins and ends using language [25, 58]. For instance, advanced Vid-LLMs like Qwen2-VL [66], while excelling at video comprehension, can struggle with grounding specific events in time. When asked, e.g., to locate “*when does the woman eat food*” in a 10-frame video, the model can hallucinate an illogical answer like “*from frame 000 to 580*.”* This limitation arises because these models are primarily trained to align visual content with language descriptions (what happens) while lacking mechanisms to directly interpret the temporal boundaries (when does it happen).[‡]

This gap in powerful Vid-LLMs leads us to think: How can we empower Vid-LLMs to extract temporal cues directly through visual recognition? A familiar human experience – flipping manga – provides an intuitive solution. When flipping manga, each numbered panel guides readers to follow the sequence of the narrative, linking visual content with a clearly defined timeline. Inspired by this, we introduce Number-Prompt (NumPro), which places unique numerical identifiers on each video frame, similar to manga panel numbers.

With NumPro, VTG is as intuitive as flipping manga. As shown in Figure 1, NumPro augments each frame with a unique numerical identifier denoting its position in the temporal sequence. Given a language query targeting an event, Vid-LLMs retrieve relevant visual features of video frames and associate them with the frame numbers overlaid. These numerical identifiers are then directly translated into textual outputs. In practice, we strategically position frame numbers in the bottom-right corner, using a defined font size and distinct color. This design ensures numbers visibility without obstructing essential visual content. Overall, NumPro allows Vid-LLMs to “read” the video timeline, effectively converting visual recognition into a temporal narrative.

NumPro’s elegance lies in its simplicity: by subtly adding frame numbers as temporal markers into video frames, we enable Vid-LLMs to naturally correlate each frame to its temporal location in the video sequence. Unlike previous approaches [20, 25, 26, 40, 55, 58, 68], NumPro does not introduce additional tokens or modify model vocabulary to provide temporal cues, thus avoiding additional learning complexities and maintaining strong transferabil-

*See more cases in Appendix 7.1.

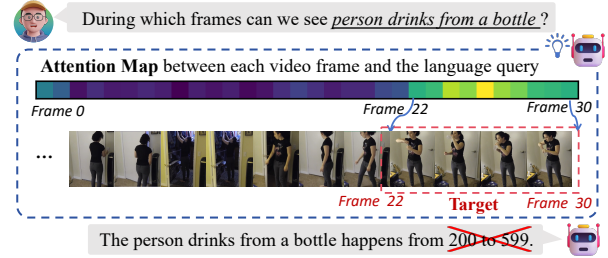


Figure 2. **Attention Analysis between Video Frames and Event Query.** Although the model accurately attends to regions of interest related to the query, it struggles to generate precise temporal boundaries in its response.

ity across various tasks and datasets. Temporal grounding, therefore, becomes an accessible, “free-lunch” enhancement for Vid-LLMs already proficient in understanding video content. Additionally, fine-tuning on a specially curated NumPro-enhanced VTG dataset (NumPro-FT) further advances state-of-the-art performance.

Our contributions can be summarized as follows:

- We introduce NumPro, a novel approach that enhances Video Temporal Grounding (VTG) capabilities of Vid-LLMs by overlaying frame numbers onto video frames, making temporal grounding as intuitive as following numbered panels in flipping manga.
- Through an experimental study, we find a suitable NumPro design (font size, color, and position) that ensures high detectability by the model while minimally interfering with the original video content.
- We thoroughly evaluate NumPro on standard VTG benchmarks and metrics in both training-free and fine-tuned scenarios, demonstrating its effectiveness across various models and datasets.

2. Related Work

Video Temporal Grounding with Vid-LLMs. Video Temporal Grounding (VTG) [39, 52, 65, 90] focuses on the precise identification of event timestamps within videos, covering tasks such as moment retrieval [3, 4, 7, 15, 17, 18, 34, 45, 64, 75, 76, 81–83, 86], dense captioning [4, 9, 19, 22, 28, 31, 56, 64, 67, 77, 85], and highlight detection [33, 58]. For current Video Large Language Models (Vid-LLMs) [35, 66, 89], which leverage powerful LLMs [1] for cross-modal understanding and video-based reasoning, VTG is crucial for achieving fine-grained temporal and visual comprehension, enabling end-to-end video dialogue systems with integrated temporal reasoning [25, 58, 68]. To achieve this, some methods rely on refined instruction datasets with temporal information (timestamps or frame numbers) for model fine-tuning [25, 40], while others concatenate additional textual temporal timestamps tokens with visual inputs [23, 61] or introduce spe-

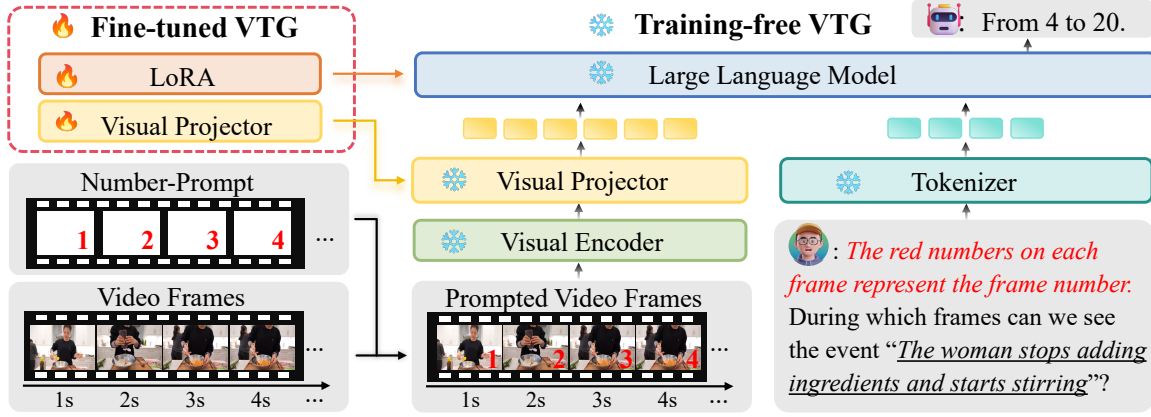


Figure 3. **Framework of Our Approach in Two Settings:** (1) Training-free VTG with NumPro, where frame numbers are directly added to video frames, enabling Vid-LLMs to locate events temporally without additional training, and (2) Fine-tuned VTG with NumPro-FT, which further improves VTG performance by fine-tuning Vid-LLMs on a dataset NumPro-enhanced with no architectural modifications.

cific temporal embeddings [20, 58]. Additional strategies model video structure [21, 26, 68] to better segment or organize videos into parts suitable for VTG. However, these approaches often require extensive retraining or specialized model adaptations, limiting their flexibility and transferability. In contrast, our NumPro aims to improve VTG for existing Vid-LLMs without additional training costs or architectural modifications.

Visual Prompt in VLMs. Visual prompts [70], taking various forms such as circles [5, 78], bounding boxes [11, 14, 47] and semantic masks [49, 80], enhance vision-language models (VLMs) [48, 62, 69, 72, 79, 88] to focus on and reason about specific visual regions and reduce the occurrence of hallucination [5]. For CLIP [57], a simple red circle [60] or colored region [80] can effectively guide model attention. Multi-modal large language models (MLLMs) [2, 6, 38] are also sensitive to specific visual prompts [5]. For example, ViP-LLaVA [5] and SoM [78] prompt MLLMs to answer about specific image regions with graphic shapes or numeric tags. CoLLaVO [32] and DOrA [71] utilize pixel-level prompts in images or videos to enhance the semantic localization capability of MLLMs. Additionally, toolchain [59, 63, 73, 92] approaches aggregate various visual prompts into multi-step reasoning paradigm to support reasoning complex tasks. While prior works focus on enhancing the region-based visual understanding of VLMs with visual prompts, our NumPro is the first to employ simple numerical tags as visual prompts within video frames to improve the temporal grounding capability.

3. Number-Prompt Approach

Our Number-Prompt (NumPro) approach provides a simple yet effective solution to enhance Video Temporal Grounding (VTG) capabilities of existing Video Large Language Models (Vid-LLMs) in both training-free and fine-tuned

settings, as shown in Figure 3. Section 3.1 presents an attention analysis based on Qwen2-VL [66] to highlight the challenge of aligning visual features with textual temporal boundaries. Section 3.2 describes the construction of NumPro and the fine-tuning process of Vid-LLMs on a NumPro-augmented VTG dataset, referred to as NumPro-FT. Finally, Section 3.3 details the design optimization of NumPro for maximizing its effectiveness.

3.1. Attention Analysis

Current Vid-LLMs process videos as a sequence of frames. Visual representations of the video can be taken as the concatenated representations from each individual frame, aggregating the information from discrete frames into a comprehensive video level. This allows Vid-LLMs to understand videos by aligning visual representations of frame images with the textual representations of language queries.

To explore the challenge in video temporal grounding (VTG), we analyze the attention map between representations of the frame image tokens and the query language tokens, and then we assess the temporal description of relevant video frames. Using Qwen2-VL-7B [66] as a case study, we highlight the challenge of VTG for Vid-LLMs: while Vid-LLMs can understand what event is happening within a video, they struggle to translate this understanding into a textual description that describes when the event begins and ends.

Specifically, we take a video and a language query as input, and extract the attention scores from the final multi-head self-attention layer of Qwen2-VL-7B [66]. For each frame within the video sequence, we aggregate the attention scores from all the visual tokens corresponding to that frame across all attention heads. As illustrated in Figure 2, the attention map reveals a strong correlation between the text query of an event and targeted video segments. It in-

indicates that Qwen2-VL-7B can effectively focus on query-relevant frames, which is consistent with the model’s strong performance in other content-related video understanding tasks [16, 37, 74]. However, the model struggles to verbalize the correct temporal boundaries, and generates surprising hallucinations such as “from 200 to 599.”[†]. This observation underscores the need for mechanisms that can bridge the gap between spatial feature alignment and temporal reasoning with Vid-LLMs, which we aim to address.

3.2. NumPro and NumPro-FT

Our approach, Number-Prompt (NumPro), empowers Vid-LLMs to directly associate specific visual content with its temporal information, turning temporal localization into a visual alignment task. As shown in Figure 3, NumPro operates in both training-free and fine-tuned scenarios.

In the training-free setting, each video frame is marked with its corresponding frame number. By utilizing the built-in Optical Character Recognition (OCR) capabilities of Vid-LLMs, we enable them to “read” the timeline through the frame numbers associated with visual content. To clarify the purpose of the added numbers to Vid-LLMs, we prepend a simple instruction to each event query: “The red numbers on each frame represent the frame number.” This approach allows Vid-LLMs to identify frame-level boundaries by directly linking the frame numbers to language queries.

For improved performance, NumPro-FT fine-tunes Vid-LLMs on a NumPro-augmented dataset. This stage aligns frame numbers with temporal spans within the training data, embedding temporal grounding capabilities into the model’s learned representations. During fine-tuning, we freeze the visual encoder and only fine-tune the visual projector and LLM components. To reduce parameter count and training overhead, we apply Low-Rank Adaptation (LoRA) [24] to adjust the LLM. Our training objective is to maximize the likelihood of generating the correct answer tokens $\mathbf{A}$ via auto-regressive language modeling:

$$P(\mathbf{A} \mid V, T_{\text{instruct}}) = \prod_{j=1}^L P_{\theta}(A_j \mid V, X_{\text{instruct}}, \mathbf{A}_{<j}) \quad (1)$$

Here, V represents the input video, θ denotes the trainable parameters, T_{instruct} is the text instruction, L is the length of the answer sequence $\mathbf{A}$, and $\mathbf{A}_{<j}$ includes all preceding answer tokens before the current token A_j .

3.3. Design of Numerical Prompt

An effective NumPro design must ensure: (1) numbers are easily recognized by the model, and (2) minimal interference with visual content. Previous research [5] indicates that the appearance and placement of visual prompts can

[†]See more cases in Appendix 8

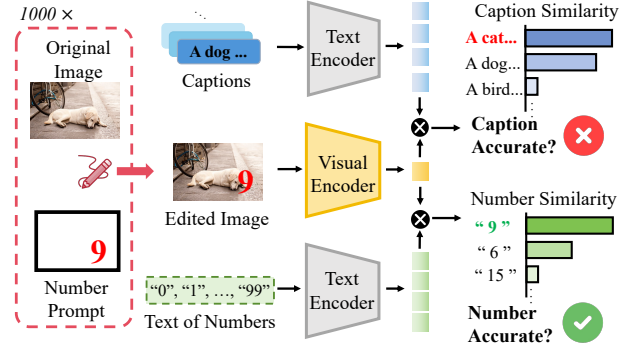


Figure 4. **Our NumPro Design Algorithm.** We overlay different numbers onto COCO images and obtain visual and textual representations using CLIP encoders. For each configuration, we calculate Number/Caption Similarity and derive Number/Caption Accuracy, to identify the optimal NumPro design that balances recognizability and minimal disruption to the visual content.

influence model attention. Given that all Vid-LLMs operate at a fixed resolution of 336×336 , we optimize NumPro by assessing three factors: font size, color, and placement position of the frame number.

To determine an effective NumPro design, we use two primary metrics: Number Accuracy, assessing how well the model identifies overlaid numbers, and Caption Accuracy, measuring how accurately the original caption aligns with frame content after adding numbers. Balancing these two metrics allows us to select NumPro configurations where the numbers are clearly recognizable without disrupting the main video content.

To make the design choices robust across various models and datasets, we employ CLIP-based experiments on a subset of MSCOCO dataset [42] to calculate Number Accuracy and Caption Accuracy separately. We use the CLIP ViT-B/32 [8, 12, 27, 30, 46, 57, 91] model to generate visual and textual representations, as many Vid-LLMs utilize CLIP-style vision encoders [13, 44, 66], allowing our findings to generalize well across Vid-LLMs. COCO image-caption pairs serve as proxies for video frames, avoiding the high costs and limited scalability of direct VTG testing. Specifically, we randomly select 1,000 distinct image-caption pairs from MSCOCO [42] and overlay numbers ranging from “0” to “99” onto the image in various configurations.

As shown in Figure 4, we first obtain representations from CLIP [57] vision and text encoders and compute intermediate similarity scores (i.e., Number and Caption Similarity) between them. Using the added numbers and original captions as ground truth, we select the text numbers and captions with the highest similarity scores as predictions to calculate Number and Caption Accuracy. Configurations balancing these accuracies are optimal for NumPro design.

As shown in Figure 5, our findings indicate that increas-

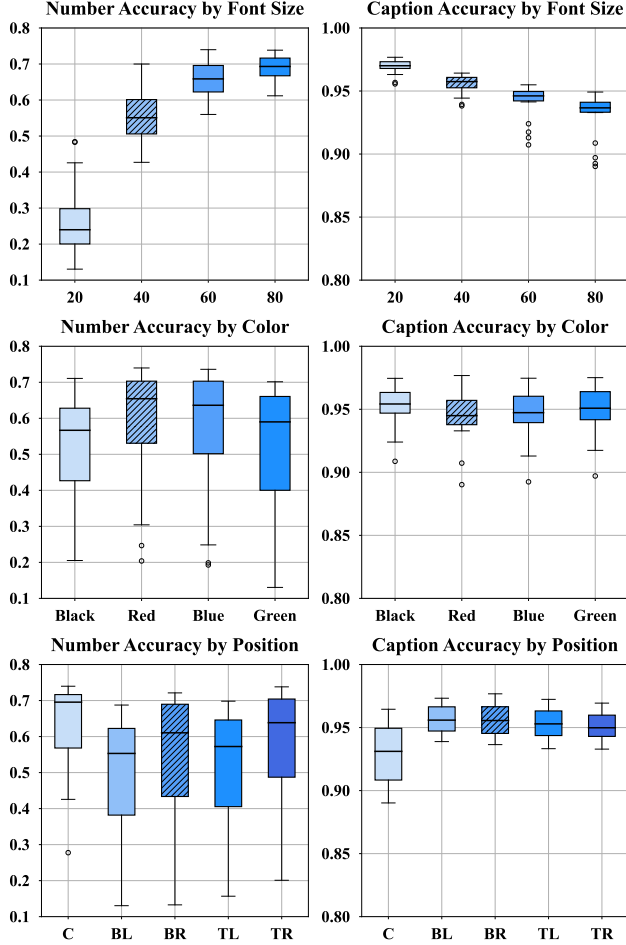


Figure 5. **The Impact of Different Number-Prompt Designs.** We categorize the design into three dimensions: font size, position, and color. BL stands for Bottom Left, BR for Bottom Right, TL for Top Left, TR for Top Right, and C for Center.

ing the font size improves number accuracy but reduces caption accuracy, suggesting that a moderate font size (40 or 60) is optimal. For color selection, caption accuracy remains relatively stable across different colors. Red shows the best performance for number accuracy, while black was the least effective. This finding is also consistent with previous works [5, 60]. Additionally, positioning the text in the center of the image significantly reduced caption accuracy due to overlaps with key visual elements, while placing the numbers in the bottom-right corner provides the best balance between caption and number accuracy. Finally, we select a font size of 40, the color red, and the bottom-right position for our final NumPro design.

In practice, CLIP-based designs provide approximate rather than definitive guidance, further testing on Vid-LLMs with a VTG dataset may yield additional model-specific insights. In Sec 4.3, consistent results further validate the effectiveness of our design.

4. Experiments

We evaluate our model on two Video Temporal Grounding (VTG) tasks: Moment Retrieval [4, 18] and Highlight Detection [33]. Moment Retrieval, given a language query describing an event, identifies the specific start and end video frames of the event. We utilize Charades-STA [18] and ActivityNet [4] as evaluation datasets, following previous works [25, 55, 58, 68]. Evaluation metrics include the mean Intersection over Union (mIoU) and recall@1 at various IoU thresholds m ($R@m$), where m is set to $\{0.3, 0.5, 0.7\}$ following previous work [25, 58]. For Highlight Detection, which aims to locate and rank video frames based on their relevance to the language query, we use QVHighlights [33] for evaluation. Evaluation metrics include mean Average Precision (mAP) and HIT@1 (the hit ratio of the highest-scored clip), as in [20, 33, 58]. Please see Appendix 13 for more task examples.

4.1. Implementation Details for NumPro-FT

Dataset Preparation. Our temporal grounding dataset consists of 70k question-answer pairs from DiDeMo [51] and ActivityNet Caption [4] datasets. Additionally, we incorporate data from Stage 2 and Stage 3 of VTimeLLM dataset [25]. After filtering out invalid videos, we obtain a comprehensive instruction dataset totaling 220k samples. Each video in our dataset is augmented with our NumPro method by overlaying frame numbers directly onto the video frames. The question-answer pairs follow a consistent template: questions are formatted as “During which frames can we see {query}?” and answers are formatted as “From x to y ”, where x and y denote the start and end frame numbers of the query event.

Training Details. We utilize the LongVA-7B-DPO [87] as our base model, taking into account its uncomplicated design and its extensive capacity to handle context length. Additionally, it has not been trained on any video data. The model is trained for 3 epochs over our curated dataset with a total batch size of 128. We use the AdamW optimizer [29] with cosine learning rate decay. The learning rate is set to $1e-4$, and the warm-up ratio is 0.05. The LLM component utilizes LoRA with parameters $r = 64$ and $\alpha = 128$. All experiments are conducted on 8 H800 GPUs.

4.2. Main Results

4.2.1. Comparison with State-of-the-Art Methods

Table 1 presents a comparative analysis of Vid-LLMs enhanced with our NumPro/NumPro-FT against existing state-of-the-art (SOTA) methods on Moment Retrieval and Highlight Detection tasks.

Moment Retrieval: Applying training-free NumPro enables Vid-LLMs to approach or exceed previous SOTA performance, benefiting both closed-source and open-source

Table 1. **Comparison of performance on the video temporal grounding task with previous state-of-the-art methods.** *NumPro* refers to the use of number prompts for augmentation during inference, while *NumPro-FT* indicates fine-tuning with the number prompt augmentation instruction dataset. The best results are highlighted in **bold**, and the second-best are underlined.

Model	Charades-STA				ActivityNet				QVHighlights	
	R@0.3	R@0.5	R@0.7	mIoU	R@0.3	R@0.5	R@0.7	mIoU	mAP	HIT@1
<i>VTG-Tuned Vid-LLMs</i>										
GroundingGPT [40]	-	29.6	11.9	-	-	-	-	-	-	-
LITA [26]	-	-	-	-	-	25.9	-	28.6	-	-
VTG-LLM [20]	52.0	33.8	15.7	-	-	-	-	-	16.5	33.5
TimeChat [58]	47.7	22.9	12.5	30.6	30.2	16.9	8.2	21.8	14.5	23.9
VTimeLLM [25]	51.0	27.5	11.4	31.2	44.0	27.8	14.3	30.4	-	-
Momentor [55]	42.9	23.0	12.4	29.3	42.6	26.6	11.6	28.5	7.6	-
HawkEye [68]	50.6	31.4	14.5	33.7	<u>49.1</u>	29.3	10.7	32.7	-	-
<i>General Vid-LLMs</i>										
GPT-4o [53]	55.0	32.0	11.5	35.4	33.3	21.2	10.4	23.7	<u>39.5</u>	<u>68.7</u>
+NumPro	57.1	35.5	13.5	37.6	45.5	<u>30.8</u>	<u>18.4</u>	<u>33.6</u>	40.5	70.7
Qwen2-VL-7B [66]	8.7	5.4	2.4	7.9	17.0	9.4	3.9	12.5	21.5	42.2
+NumPro	<u>60.7</u>	<u>36.8</u>	<u>15.9</u>	<u>38.5</u>	44.2	26.4	14.4	31.3	23.6	43.4
LongVA-7B-DPO [87]	22.6	10.1	2.2	14.6	11.8	5.3	1.9	8.2	14.2	20.4
+NumPro	27.2	10.3	2.9	18.9	20.1	10.8	5.4	15.2	15.3	24.3
+NumPro-FT	63.8	42.0	20.6	41.4	55.6	37.5	20.6	38.8	25.0	37.2

Vid-LLMs. GPT-4o [53] already exhibits strong moment retrieval capabilities, and our NumPro further enhances the performance. In particular, NumPro achieves a 9.9% increase in mIoU on ActivityNet, surpassing the previous SOTA by 0.9%. Qwen2-VL-7B performs poorly initially and also sees a significant improvement with NumPro, averaging a 24.7% increase in mIoU across datasets.

Moreover, starting from a relatively low baseline on LongVA-7B-DPO [87], our fine-tuning approach, NumPro-FT, establishes new SOTA across all metrics. On Charades-STA, it surpasses previous SOTA by 11.8%, 8.2%, 4.9%, 7.7% (R@0.3, R@0.5, R@0.7, mIoU), and on ActivityNet, it surpasses previous SOTA by 6.5%, 8.2%, 6.3%, 6.1% (R@0.3, R@0.5, R@0.7, mIoU). These results demonstrate that NumPro and NumPro-FT can utilize the superior video understanding abilities of existing Vid-LLMs and significantly enhance their moment retrieval capabilities.

Highlight Detection: In this task, models like GPT-4o [53] and Qwen2-VL have already achieved state-of-the-art (SOTA) performance. However, our NumPro approach consistently enhances their performance, with an average increase of 1.55% in mean Average Precision (mAP) and 1.6% in the hit ratio of the highest-scored clip (HIT@1). Additionally, applying NumPro-FT enables LongVA-7B-DPO to surpass existing SOTA by a large margin (+8.5% in mAP and +3.7% in HIT@1). These findings suggest that NumPro and NumPro-FT, which can be easily appended to current Vid-LLMs, hold substantial potential for further ad-

vancing temporal reasoning capabilities.

4.2.2. Effectiveness of NumPro across Vid-LLMs

Beyond surpassing SOTA, Table 2 demonstrates the broad applicability and scalability of NumPro across various Vid-LLMs in Video Temporal Grounding. We apply NumPro to additional Vid-LLMs, including LLaVA-Video-7B [89], LLaVA-OneVision-7B [35], and Qwen2-VL-72B [66], and observe notable performance improvements, with average mIoU gains reaching up to 18.1% on Charades and 14.0% on ActivityNet. Moreover, we conduct fine-tuning experiments with and without NumPro-augmented data (indicated as +FT in Table 2). Results show that NumPro-FT consistently outperforms conventional fine-tuning, particularly on longer video datasets like ActivityNet, where it achieves a substantial 9.8% gain in mIoU. Additional studies on NumPro’s effectiveness for QVHighlights are provided in Appendix 10. Those observations underscore the effectiveness of NumPro across models and highlight its superior impact when combined with fine-tuning.

4.2.3. Qualitative Results

In Figure 6, we compare our method with SOTA methods, TimeChat [58] and VTimeLLM [25], through two visualization cases from our ActivityNet dataset. The first example features minimal scene changes between video frames. TimeChat predicts an early start, while VTimeLLM fails to capture the full event duration. In contrast, our method precisely captures the correct event boundaries. The sec-

Table 2. Performance of Applying NumPro to Various Vid-LLMs and Ablation Results on NumPro-FT.

Model	Charades-STA				ActivityNet			
	R@0.3	R@0.5	R@0.7	mIoU	R@0.3	R@0.5	R@0.7	mIoU
LLaVA-OneVision-7B [35]	22.3	7.9	2.1	15.9	7.1	3.1	1.1	6.1
+NumPro	42.9 (+20.6)	19.4 (+11.5)	6.6 (+4.5)	28.1 (+12.2)	14.4 (+7.3)	7.9 (+4.8)	3.8 (+2.7)	11.3 (+5.2)
LLaVA-Video-7B [89]	11.8	2.7	0.1	9.8	7.4	3.1	1.2	6.2
+NumPro	56.7 (+44.8)	25.6 (+22.9)	8.6 (+8.5)	34.6 (+24.8)	25.2 (+17.8)	15.2 (+12.1)	8.4 (+7.2)	18.6 (+12.4)
Qwen2-VL-72B [66]	0.0	0.0	0.0	0.2	1.0	0.6	0.3	1.0
+NumPro	25.8 (+25.8)	9.9 (+9.9)	3.0 (+3.0)	17.4 (+17.2)	35.5 (+34.5)	21.4 (+20.8)	11.0 (+10.7)	25.5 (+24.5)
LongVA-7B-DPO [87]	22.6	10.1	2.2	14.6	11.8	5.3	1.9	8.2
+FT	62.0	41.6	19.9	40.2	41.8	25.7	13.7	29.0
+NumPro-FT	63.8 (+41.2)	42.0 (+31.9)	20.6 (+18.4)	41.4 (+26.8)	55.6 (+43.8)	37.5 (+32.2)	20.6 (+18.7)	38.8 (+30.6)

Table 3. Ablation study on various NumPro designs. We divide the designs into three dimensions: font size, color, and position.

Size	Color	Position	Charades-STA			
			R@0.3	R@0.5	R@0.7	mIoU
40	Red	Top Left	56.7	32.9	13.8	35.8
40	Red	Top Right	58.2	34.0	13.0	36.8
40	Red	Center	53.7	29.5	10.4	34.1
40	Red	Bottom Left	61.6	37.8	15.9	39.3
40	Red	Bottom Right	<u>60.7</u>	<u>36.8</u>	<u>15.9</u>	<u>38.5</u>
20	Red	Bottom Right	53.6	34.0	14.0	34.6
40	Red	Bottom Right	60.7	36.8	15.9	38.5
60	Red	Bottom Right	<u>58.0</u>	<u>34.5</u>	<u>14.1</u>	<u>37.1</u>
80	Red	Bottom Right	<u>58.0</u>	33.9	13.7	36.9
40	Red	Bottom Right	60.7	36.8	15.9	38.5
40	Blue	Bottom Right	<u>57.8</u>	34.2	<u>14.6</u>	<u>36.6</u>
40	Black	Bottom Right	56.6	<u>36.0</u>	15.9	<u>36.6</u>
40	Green	Bottom Right	56.0	33.8	14.5	36.0

ond case involves a shorter event duration and frequent scene changes. TimeChat completely misses the event, and VTimeLLM overestimates the event duration by including irrelevant segments. Our approach, again, precisely delineates the event boundaries. These qualitative examples underscore the robustness and precision of our method in scenarios that are especially challenging for other SOTA methods. We provide additional cases on moment retrieval and highlight detection in Appendix 13.

4.3. Validation of NumPro Design

Following our heuristic design process in Sec 3.3, we validate its effectiveness in temporal grounding tasks to confirm that these design choices generalize beyond the COCO dataset. We conduct moment retrieval experiments on Charades-STA [18] with Qwen2-VL-7B [66] in a training-free setting. As shown in Table 3, the results align closely with our initial observations from the COCO dataset, confirming the effectiveness of our design choices in VTG tasks. Specifically, (1) Position: Consistent with our CLIP-based findings, placing the text in the center has the largest

impact on performance due to overlaps, while our choice of the bottom-right performs comparably to the best position; (2) Font Size: Both very large and very small fonts yield suboptimal results, supporting our balanced selection; (3) The performance on VTG is sensitive to number color, yet the red color consistently delivers the best performance, which may attribute to its high contrast against typical backgrounds [60]. Overall, the alignment between the CLIP-based design choices and the VTG results shows the validity and robustness of our NumPro design. Please refer to Appendix 11 for the ablation results of NumPro-FT. We also try directly overlaying timestamps (*e.g.*, “10.5s”) on frames, which show inferior performance than frame numbers (Appendix 12).

4.4. Investigation on the Sampling of NumPro

Typically, we augment every frame in a video with NumPro. In this section, we evaluate the impact of varying the sampling ratio and sampling method (randomly or uniformly) when selecting a subset of frames from the video to augment NumPro. As depicted in Figure 7, performance increases with more labeled frames, with uniform sampling generally maintaining higher accuracy. Notably, labeling just 20% of the frames provides a substantial performance boost and uniform sampling of 80% of the frames surpasses previous state-of-the-art, underscoring the robustness of our NumPro approach.

4.5. Influence on General Video-QA

To explore the broader applicability of NumPro, we integrate it into general video-QA tasks, using VideoInstruct [50] as our benchmark. As detailed in Table 4, the incorporation of NumPro minimally affects general comprehension metrics, with a slight decrease in Distraction Overlap (DO, -0.02) and an enhancement in Temporal Understanding (TU, +0.1). This indicates that Vid-LLMs equipped with NumPro maintain robust performance in general video-QA while excelling in precise video temporal grounding (VTG) tasks. This dual capability allows us

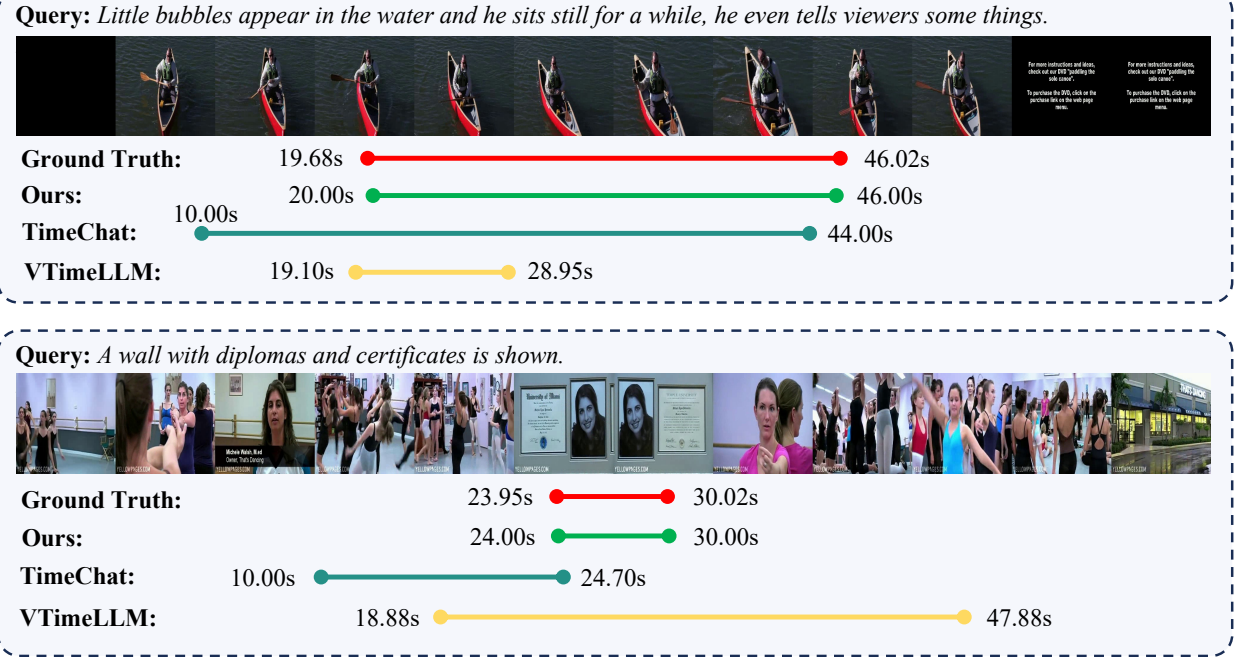


Figure 6. **Qualitative Comparison with State-of-the-Art.** Our LongVA-7B-DPO model, fine-tuned with NumPro-FT, outperforms TimeChat [58] and VTimeLLM [25] on ActivityNet by accurately identifying event boundaries in challenging scenes.

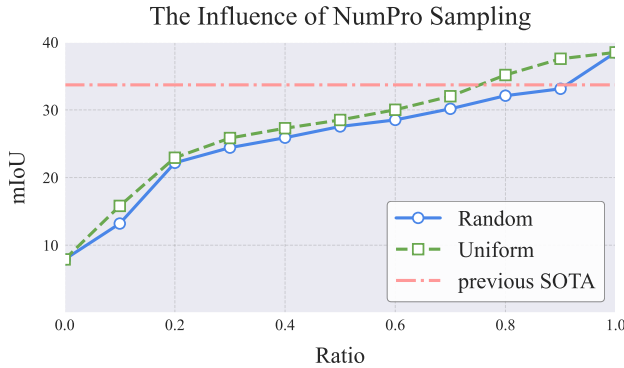


Figure 7. **Performance Comparison of Sampling Strategies for NumPro.** We compare the effects of NumPro with different sampling ratios and sampling methods (random vs. uniform), as tested on the Charades-STA [18] using the Qwen2-VL-7B [66] model.

to harness a powerful Vid-LLM for end-to-end video understanding that can flexibly adapt to both general and temporally nuanced questions within conversational AI systems. Moreover, we examine NumPro on more video-QA benchmarks including MVBench [37] and VideoMME [16], and we show Vid-LLMs enhanced with NumPro achieve robust performance across a variety of downstream tasks. Details can be found in Appendix 9.

5. Conclusion

In this paper, we propose Number-Prompt (NumPro), a simple yet efficient visual prompt designed to en-

Table 4. **The influence of applying NumPro to general video-QA.** CI stands for correctness of information, DO stands for detail orientation, CU stands for contextual understanding, TU stands for temporal understanding, and CO stands for consistency.

Model	CI	DO	CU	TU	CO
Qwen2-VL	3.10	2.57	3.46	2.47	3.30
+NumPro	3.10	2.55	3.46	2.57	3.30

hance the video temporal grounding (VTG) capabilities of Video Large Language Models (Vid-LLMs) with no effort. By overlaying frame numbers onto video content, NumPro leverages the inherent Optical Character Recognition (OCR) and visual-language alignment capabilities of Vid-LLMs, allowing them to accurately map events to specific temporal boundaries. Through systematic design informed by COCO-based heuristics and validated across VTG benchmarks, we demonstrated that NumPro effectively supports fine-grained temporal understanding while preserving general video comprehension. Through extensive evaluations, we demonstrated that NumPro consistently achieves state-of-the-art performance in both training-free and fine-tuned settings, enabling adaptable integration into both closed-source and open-source Vid-LLMs. NumPro-FT further refines temporal grounding performance, establishing new SOTA across VTG tasks. Besides, the minimal impact on general video-QA shows that NumPro can augment VTG while maintaining robust video understanding.

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